

WavInWav: Time-Domain Speech Hiding via Invertible Neural Network

Wei Fan¹, Kejiang Chen², *Senior Member, IEEE*, Xiangkun Wang³, Weiming Zhang⁴, and Nenghai Yu⁵

Abstract—Data hiding is essential for secure communication across digital media, and recent advances in Deep Neural Networks (DNNs) provide enhanced methods for embedding secret information effectively. However, previous audio hiding methods often result in unsatisfactory quality when recovering secret audio, due to their inherent limitations in the modeling of time-frequency relationships. In this paper, we explore these limitations and introduce a new DNN-based approach. We use a flow-based invertible neural network to establish a direct link between stego audio, cover audio, and secret audio, enhancing the reversibility of embedding and extracting messages. To address common issues from time-frequency transformations that degrade secret audio quality during recovery, we implement a time-frequency loss on the time-domain signal. This approach not only retains the benefits of time-frequency constraints but also enhances the reversibility of message recovery, which is vital for practical applications. We also add an encryption technique to protect the hidden data from unauthorized access. Experimental results on the VCTK and LibriSpeech datasets demonstrate that our method outperforms previous approaches in terms of subjective and objective metrics and exhibits robustness to various types of noise, suggesting its utility in targeted secure communication scenarios.

Index Terms—Time-domain audio hiding, invertible neural networks.

I. INTRODUCTION

DATA hiding is the science and technology of embedding secret information into a cover object without perceptibly altering it. During the hiding process, a secret message is concealed within a cover message in an imperceptible manner. During the revealing process, the secret message is extracted from the stego message. With the increasing popularity of digital media, such as digital audio [1], [2], [3], images [4], [5], and videos [6], these media forms are widely employed as cover objects in data hiding applications.

For digital audio, numerous internet communication applications (e.g., iTunes, YouTube, Twitter) provide audio storage and communication services, providing a diverse range of digital audio cover objects. Furthermore, leading music-sharing platforms

such as Spotify,¹ SoundCloud,² and Bandcamp³ offer additional opportunities for implementing audio data hiding techniques. These platforms collectively broaden the spectrum of available channels for audio data hiding, creating an environment conducive to the secure transmission of hidden messages under the guise of sharing audio. Traditional data hiding techniques in audio often involve manually modifying audio data using rules such as altering the least significant bit (LSB) of the audio [7], [8], [9], [10]. The domains involved in these traditional methods encompass not only the commonly used time domain but also various transform domains and compressed domains [11], [12], [13], [14], [15], [16], [17], [18]. However, these traditional methods have limited capacity (for instance, LSB methods typically use only one or two bits per sample), making them unsuitable for concealing higher information payloads such as images or audio [6]. Recently, researchers have employed neural networks as the function for audio data hiding, enabling the concealment of multimedia data within audio. Given the sensitivity of the Human Auditory System (HAS) to audio frequencies, modeling time-frequency relationship of the audio data becomes crucial. Therefore, references [2], [19], [20] hide image data within the time-frequency-domain data obtained after performing Short-Time Fourier Transform (STFT) on the audio data. STFT transforms the audio into a complex matrix consisting of the Fourier transforms of different time frames, and its inverse transform reconstructs the time-domain waveform from the complex matrix. Since STFT transformation introduces distortion, Kreuk et al. [3] incorporate differentiable STFT layers into the encoder and decoder to facilitate the learning of embedding and extraction processes in the presence of communication channel distortion. Their approach effectively conceals one audio segment within another.

However, previous works on hiding audio within audio primarily focus on the quality of the stego audio while giving less consideration to the quality of the recovered secret audio. Specifically, they recover the amplitude spectrum of the secret audio from the stego audio and reconstruct the secret audio relying on the inferred amplitude spectrum. But this reconstruction process is far from perfect, as the amplitude spectrum of the secret audio recovered from a distorted channel is invalid. In reality, there does not exist a reasonable audio with the same invalid amplitude

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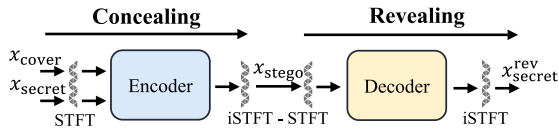
The authors are with the Anhui Province Key Laboratory of Digital Security, University of Science and Technology of China, Hefei 230026, China (e-mail: chenkj@ustc.edu.cn).

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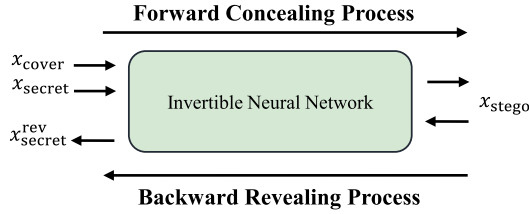
¹<https://open.spotify.com/>

²<https://soundcloud.com/>

³<https://bandcamp.com/>



(a) Traditional framework relying on spectrograms.



(b) Our flow-based framework with time-domain hiding.

Fig. 1. Comparison between framework relying on spectrograms and our flow-based framework with time-domain hiding.

spectrum, leading to a lower quality of the reconstructed secret audio.

Considering the advantages of directly using speech audio as the secret message, such as preserving speaker characteristics, intonation, and rhythm for subsequent recognition and authentication, these properties are useful only if they are still perceptible in the recovered secret in real secure communication scenarios. Therefore, the quality of the message audio is crucial for audio hiding. However, the low-quality secret audio from previous methods may prevent these advantages from being fully realized. To address the issue of previous methods relying on distorted spectrograms during secret reconstruction, we propose directly hiding the message in the time domain, which avoids reconstructing the secret from distorted spectrograms while still utilizing time-frequency constraints to preserve the advantages of modeling the time-frequency relationship. Specifically, we employ a flow-based invertible network to jointly model the hiding and revealing processes. In this architecture, the forward pass performs hiding and the inverse pass performs revealing, enhancing the reversibility of message embedding and extraction compared with encoder-decoder architectures. The distinction between the proposed method and previous approaches is illustrated in Fig. 1.

Building upon this framework, and in response to the challenges posed by lossy channels, we introduce a noise layer featuring a range of common differentiable noises during our training phase. This enhancement allows the model to efficiently process the extraction of secret audio even in suboptimal channel conditions, ensuring consistently high-quality recovery across various scenarios. Moreover, to safeguard the secret audio from unauthorized access, we incorporate an encryption-decryption module within our system. Secret audio can only be decrypted with a matching decryption key, significantly enhancing system security. Experiments, both subjective and objective, conducted on the VCTK and LibriSpeech datasets demonstrate the significant superiority of our method in terms of stego and recovered secret quality compared to previous approaches. Furthermore, our method exhibits robustness against various common distortions.

Our contributions can be summarized as follows:

- To the best of our knowledge, we are the first to explore the hiding and subsequently end-to-end recovering of a complete secret audio segment within a cover audio. Unlike previous methods that reconstructed secret audio from the spectrum, we model the hiding and revealing processes directly in the time domain, using time-frequency loss as a constraint to avoid distortion during time-frequency transformation, thus enhancing the quality of the secret audio.
- We employ a novel invertible network that simultaneously trains the hiding and revealing processes. Compared to traditional encoder-decoder architectures, our introduced invertible network exhibits better reversibility in the message embedding and extraction processes.
- We use a carefully designed composite noise layer to enhance the robustness of the invertible network. Additionally, to protect the secret messages from unauthorized extraction, we introduce a simple yet effective encryption-decryption module, enhancing the overall security of the system.

The remaining sections of this paper are organized as follows. Section II conducts a review of related works. Section III introduces our proposed method in detail. Section IV gives the experimental results and discussions, while Section V serves as the conclusion to our work.

II. RELATED WORKS

A. Traditional Audio Data Hiding

Traditionally, audio data hiding leverages the real or perceived redundancies within cover signals, employing techniques across various domains like time domain, transform domain, and compressed domain.

In the time domain, one of the earliest data hiding methods embed secret information into least significant bits (LSB) of each cover sample. References [7], [8], [9], [21] encrypt the secret into ciphertext and utilize the LSB method to embed it into cover audio to enhance security. Nassrullah et al. [10] propose an adaptive audio data hiding strategy based on LSB, dynamically balancing distortion rates and embedding capacity as required. Chen et al. [22] employ psychoacoustic models alongside adversarial examples, enhancing both the imperceptibility and undetectability of traditional LSB techniques. These LSB methods offer relatively high hiding capacities but are vulnerable to modification. Furthermore, for multimedia data such as images or audio, the hiding capacity of LSB (one or two bits per sample) remains limited. Other time-domain data hiding methods include phase coding, echo hiding, and spreading, considering the characteristics of the HAS to hide information in areas imperceptible to humans. Djebbar et al. [23] use the low sensitivity of HAS to phase distortion to hide information by altering audio phase values, providing a degree of robustness. Oh et al. [24] utilize the temporal masking effect of HAS, introducing slightly delayed echo signals into cover audio, rendering them robust against common signal processing attacks. Matsuoka et al. [25] expand narrowband secret information across

wider frequency ranges for information hiding. These non-LSB techniques are more robust to modifications and compression but offer lower capacities.

Beyond the time domain, discrete cosine transform (DCT) [12], fast Fourier transform (FFT) [11], and discrete wavelet transform (DWT) [13] are employed in audio data hiding. Additionally, considering audio compression during transmission, particularly through MP3 and VoIP, references [14], [15], [16], [17], [18] directly embed within the compressed domain to alleviate potential losses incurred during the compression process. Related traditional data hiding methods have also been extensively studied in the image domain [26], [27], [28], [29], [30], [31], [32], [33].

In general, the capacity of traditional audio data hiding methods is far from sufficient for multimedia data. Differing from traditional methods, we employ deep neural networks to achieve high perceptual transparency and increased hiding capacity while maintaining a certain level of robustness.

B. Deep-Learning-Based Audio Data Hiding

In recent years, several deep-learning-based data hiding schemes have been proposed. Initially, there was a proposal to train neural networks for hiding images within other images [4]. In this approach, an encoder network is utilized to conceal a secret image within a cover image to generate a stego image, while a decoder network is employed to recover the secret image from the stego image. Zhu et al. [5] extend this work by introducing adversarial loss terms and noise layers, enhancing robustness against various forms of distortion.

Given that the HAS is more sensitive than the Human Visual System (HVS) [34], digital audio data hiding is typically more challenging than image hiding. Kreuk et al. [3] point out that steganographic models designed for visual content might not be suitable for audio. They suggest integrating differentiable STFT layers between the encoder and decoder to adapt the spectrogram of audio to the framework of image hiding. While this method effectively conceals and recovers identifiable secret audio, the quality of the recovered audio is compromised. Extending beyond audio, images and videos can also serve as hidden information for audio data hiding. Yang et al. [6] suggest using the reversibility of flow-based models to hide video frames within audio. References [2], [19], [20] conceal images within the audio spectrogram. Chen et al. [1] introduce a provably secure steganographic method maintaining distribution using audio generation models such as WaveGlow [35] and WaveNet [36]. Furthermore, Takahashi et al. [37] propose a robust audio hiding method for instrument source separation, concealing byte-level information within different musical instrument tracks. Previous studies have demonstrated the immense potential of DNNs in audio data hiding, however, their performance in extracting audio-type secret information remains unsatisfactory.

C. Normalizing Flow-Based Model

Normalizing flow-based models exhibit a strong ability to directly capture complex probability distributions from data. The structure of a normalizing flow-based model is reversible, allowing it to efficiently perform both forward and inverse

mappings using the same network and parameters. The property of reversible mappings makes the model particularly suitable for implementations as an Invertible Neural Network (INN). For instance, given a variable x and a forward function $y = f_{\theta}(x)$, x can be recovered through $x = f_{\theta}^{-1}(y)$, where f_{θ}^{-1} and f_{θ} share the parameters θ .

Normalizing flow-based models were initially introduced for tasks involving image generation. Pioneering work by Dinh et al. [38] first demonstrate the strong generative capabilities of normalizing flow-based models. Subsequently, Kingma et al. [39] enhance their utility in realistic image synthesis and manipulation by introducing invertible 1×1 convolutions in Glow model. Owing to their exceptional performance, flow-based models are applied in various image-related tasks. For example, Ardizzone et al. [40] introduce conditional flow-based models for guiding image generation and coloring, addressing the task of natural image generation guided by a conditioning input. Xiao et al. [41] leverage the bidirectional transformation capabilities of these models for image super-resolution, mapping between low and high-resolution images.

In the specialized area of data hiding, flow-based models demonstrate considerable efficacy. Jing et al. [42] incorporate flow-based models into image hiding, exploring the potential of modeling image revealing as the inverse process of image concealing within an invertible network architecture. Following this, flow-based models are increasingly adopted in the field, with numerous studies affirming their superiority in handling large-capacity concealment tasks [43], [44]. Furthermore, Xu et al. [45] introduce the conditional flow and noise layer into the invertible framework, thereby achieving significant improvements in robustness. Ma et al. [46] propose an image watermarking network that combines invertible and non-invertible frameworks, utilizing the non-invertible part for handling quantization noise, effectively balancing the fidelity and robustness of the watermark. These enhancements in robustness showcase the resilience of flow-based models in concealing tasks. Overall, the success of these diverse implementations indicates the broad potential of flow-based models for applications requiring precise, bidirectional transformations.

While INNs have found numerous applications in image processing tasks, to the best of our knowledge, they have not been explored in the context of audio hiding. Given the reversible nature of flow-based models, which allows for precise, bidirectional transformations, this paper presents our initial attempt to model audio concealing and revealing as a pair of inverse processes using normalizing flow-based models.

III. THE PROPOSED METHOD

This section presents the proposed novel audio data hiding approach referred to as “WavInWav”, which involves hiding secret audio within a high-capacity and imperceptible cover audio. The core concept behind this approach lies in utilizing an invertible network to concurrently model the concealing and revealing processes of time-domain data while employing time-frequency constraints as training constraints. Additionally, to enhance the robustness of the proposed method against channel distortions, we incorporate a composite noise layer

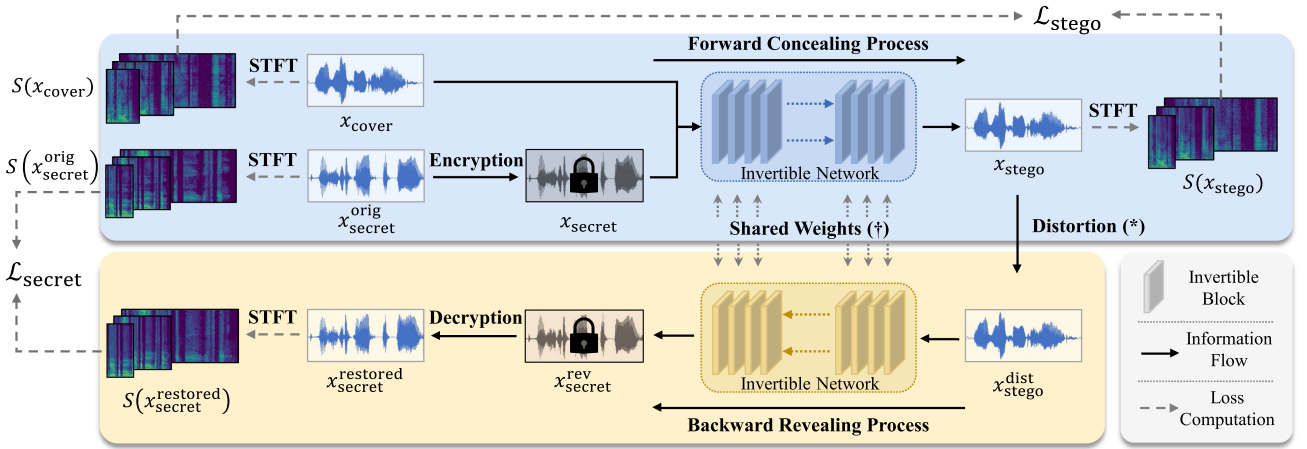


Fig. 2. Overview of the proposed audio hiding method. The audio is hidden in the time domain, while loss is computed in the time-frequency domain. The asterisk (*) indicates the distortion module active during robust training. The dagger (†) denotes default parameter sharing, except in the robust setting (*) where forward and backward are decoupled.

during the training process to simulate distortions. To further secure the hidden data against unauthorized extraction, we also integrate an encryption mechanism into our system. We outline the entire system architecture in Section III-A, explain the concept of time-domain hiding in Section III-B, introduce the network architecture we employ in Section III-C. Following this, Section III-D is dedicated to detailing the implemented security mechanisms, including the specifics of the encryption and decryption processes. Lastly, we present our chosen loss functions in Section III-E.

A. Overview

Fig. 2 illustrates the overall framework of the proposed WavInWav which utilizes a flow-based invertible neural network, referred to as WavINN. In the concealing phase, the secret audio x_{secret}^{orig} undergoes encryption to form x_{secret} . Subsequently, both the cover audio x_{cover} and the encrypted secret audio x_{secret} are input into WavINN to produce x_{stego} . This x_{stego} maintains perceptual similarity to x_{cover} while encapsulating x_{secret} within.

The reveal process utilizes the same network architecture as the concealment phase, with an identical parameter set, but reverses the information flow. During this backward phase, WavINN processes the stego audio x_{stego} to extract the recovered secret x_{secret}^{rev} and the cover audio x_{cover}^{rev} . Subsequently, the decrypted secret audio $x_{secret}^{restored}$ is reconstructed from x_{secret}^{rev} , completing the secure recovery process. Table I lists the notations used in this paper.

B. Time-Domain Audio Hiding

In this section, we provide a detailed explanation of the concept of hiding messages in the time domain. Spectrograms offer an intuitive representation of the frequency spectrum of an audio signal, which is more aligned with human auditory perception compared to the time-domain representation of audio signals. Therefore, previous methods often operate in the frequency domain, requiring the transformation of audio data into spectral data using the STFT for concealing in the frequency domain

TABLE I
NOTATIONS USED IN THE WAVINWAV FRAMEWORK

Notation	Description
x_{secret}^{orig}	Original secret audio before encryption
x_{secret}	Encrypted secret audio
x_{cover}	Cover audio used to conceal the secret
x_{stego}	Stego audio containing the encrypted secret audio
x_{secret}^{rev}	Recovered secret from the stego audio
x_{cover}^{rev}	Recovered cover audio from the stego audio
$x_{secret}^{restored}$	Decrypted secret audio reconstructed from x_{secret}^{rev}
r	Discarded information in the concealing process
z	Auxiliary variable used in the revealing process

and subsequent inverse Short-Time Fourier Transform (iSTFT) for conversion back to the time domain. Specifically, for the cover audio x_{cover} and the secret audio x_{secret} , we denote the frequency-domain hiding function as $E_f(\cdot)$ and the function for extracting secret messages from the frequency domain as $D_f(\cdot)$, with STFT denoted as $S(\cdot)$ and iSTFT denoted as $S^{-1}(\cdot)$. The forward concealment process in the frequency domain can be expressed as:

$$S^{-1}(E_f(S(x_{cover}), S(x_{secret}))) = x_{stego}. \quad (1)$$

Subsequently, the process of extracting the secret audio from the stego audio can be represented as:

$$S^{-1}(D_f(S(x_{stego}))) = x_{secret}^{rev}. \quad (2)$$

This approach is advantageous in maintaining perceptual consistency between the cover audio and the stego audio. However, it falls short in delivering satisfactory secret audio quality due to two inherent flaws. First, because the human auditory system is more sensitive to the amplitude spectrum of audio and phase spectrum are often challenging to learn, frequency-domain audio hiding typically operates on amplitude spectrum, leading to the loss of the original phase of the secret audio during the hiding process. The recovery of the secret audio phase must be inferred from the amplitude spectrum $D_f(S(x_{stego}))$. Furthermore, the

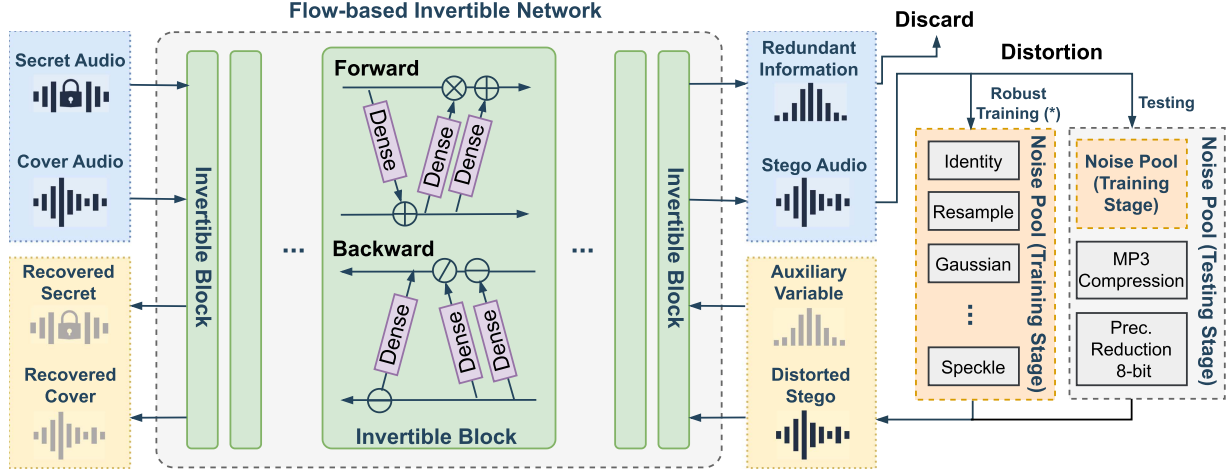


Fig. 3. The structure of the invertible neural network and noise layers used in our approach. The asterisk (*) marks the distortion module active during robust training. The forward and backward processes share the same parameters by default, except in the robust setting (*) where they are decoupled.

STFT-iSTFT transformation process is lossy for the stego audio, meaning that $S^{-1}(S(x_{\text{stego}})) \neq x_{\text{stego}}$. Consequently, the amplitude spectrum $D_f(S(x_{\text{stego}}))$ of the recovered secret audio is also lossy. This indicates that the recovered amplitude spectrum $D_f(S(x_{\text{stego}}))$ might not align with any realistic audio signal $x_{\text{secret}}^{\text{rev}}$ that satisfies the criteria of (2). In other words, a realistic audio that matches $D_f(S(x_{\text{stego}}))$ may not actually exist. This further contributes to the error between $x_{\text{secret}}^{\text{rev}}$ and x_{secret} , rendering the previous method unsatisfactory in terms of secret audio quality.

To address these two shortcomings and achieve the recovery of high-quality secret audio, we consider directly concealing information in the time domain. Fig. 2 demonstrates our concept of hiding the message in the time domain. We denote the time-domain hiding function as $E_t(\cdot)$ and the function for extracting secret messages in the time domain as $D_t(\cdot)$. The forward concealment process in the time domain can be expressed as:

$$E_t(x_{\text{cover}}, x_{\text{secret}}) = x_{\text{stego}}, \quad (3)$$

where x_{stego} represents the stego audio. The process of recovering the secret audio from the stego audio can be represented as:

$$D_t(x_{\text{stego}}) = x_{\text{secret}}^{\text{rev}}, \quad (4)$$

where $x_{\text{secret}}^{\text{rev}}$ denotes the revealed secret.

By directly concealing information in the time domain, we circumvent the need for STFT during audio hiding and extraction, thus avoiding the impact of the lossy STFT-iSTFT transformation and the challenge of recovering audio from unrealistic spectrograms.

C. Network Structure

In this section, we provide a detailed overview of the architecture of the invertible neural network WavINN that we employ. Fig. 3 illustrates the structure of the invertible neural network used in our approach. By using an invertible network, encoded

features can be highly consistent with the features required by the decoder. This approach helps to limit the inclusion of redundant features in the encoding process, resulting in a more efficient and targeted representation of the necessary information. Specifically, WavINN consists of a series of entirely invertible network modules, and its forward concealment process and backward reveal process are inverse operations. Thus, we denote the forward function as f_θ , and the backward function can be represented with the same parameters θ as f_θ^{-1} .

In the forward process, f_θ takes a pair of cover audio $x_{\text{cover}} \in \mathbb{R}^{1 \times t}$ and encrypted secret x_{secret} as input and then outputs the stego audio x_{stego} along with redundant information r . For the backward decoding process, f_θ^{-1} takes the stego audio x_{stego} and an auxiliary variable z as input. After passing through a series of network modules that share parameters with the forward network, it recovers the secret $x_{\text{secret}}^{\text{rev}}$. It should be noted that the redundant information r is discarded during this process. Therefore, it is necessary to resample an auxiliary variable z from a Gaussian distribution to replace the redundant information r as part of the input during the decoding process.

1) *Invertible Blocks*: As illustrated in Fig. 4, the concealment network and the reveal network share submodules and network parameters, but the flow of information is in opposite directions. Both the concealment network and the reveal network consist of M concealment blocks. For the i -th concealment block in the forward process, given inputs x_{cover}^i and x_{secret}^i , the outputs x_{cover}^{i+1} and x_{secret}^{i+1} can be expressed as follows:

$$x_{\text{cover}}^{i+1} = x_{\text{cover}}^i + \phi(x_{\text{secret}}^i), \quad (5)$$

$$x_{\text{secret}}^{i+1} = x_{\text{secret}}^i \cdot \exp(\alpha(\rho(x_{\text{cover}}^{i+1}))) + \eta(x_{\text{cover}}^{i+1}), \quad (6)$$

where $\alpha(\cdot)$ is a sigmoid function scaled by a constant factor used as a clamp, $\cdot$ denotes the dot product operation, and $\rho(\cdot)$, $\eta(\cdot)$, and $\phi(\cdot)$ can be arbitrary functions, not necessarily invertible ones. In this context, we employ one-dimensional dense blocks as the functions $\rho(\cdot)$, $\eta(\cdot)$, and $\phi(\cdot)$. We adopt this CNN-based architecture to efficiently process raw waveforms which inherently involve long sequences. Unlike attention-based

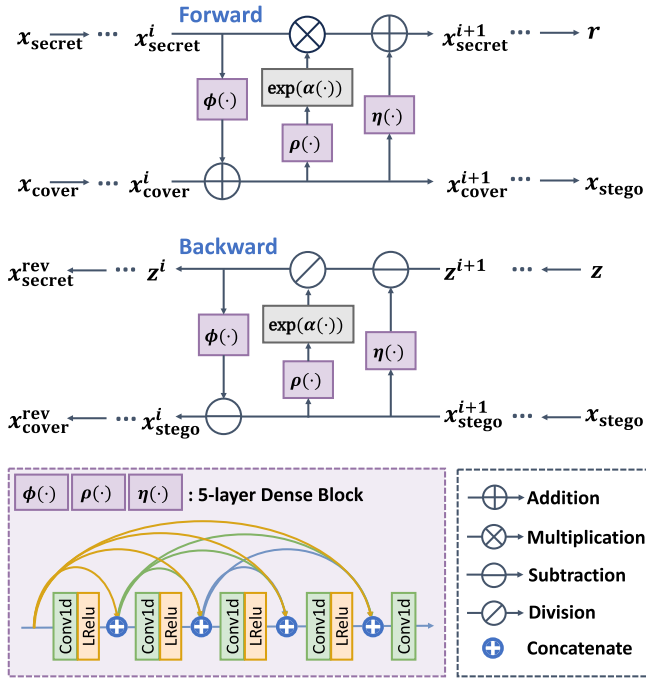


Fig. 4. The backbone of the i -th invertible block used in our approach.

models which incur high computational costs on such data, one-dimensional dense blocks offer a linear complexity and are particularly effective for modeling the local details essential for high-capacity audio hiding.

For a network with M concealment blocks, during the forward process, the first concealment block takes inputs x_{cover} and x_{secret} , while the M -th block, which is the last block, takes inputs x_{cover}^M and x_{secret}^M . The outputs are the final stego audio x_{stego} and redundant information r .

In the backward process for the i -th concealment block, the information flow is reversed compared to the forward process, going from the $(i + 1)$ -th block to the i -th block. Therefore, the inputs are x_{stego}^{i+1} and z^{i+1} , and the outputs x_{stego}^i and z^i can be expressed as follows:

$$z^i = (z^{i+1} - \eta(x_{\text{stego}}^{i+1})) \cdot \exp(-\alpha(\rho(x_{\text{stego}}^{i+1}))), \quad (7)$$

$$x_{\text{stego}}^i = x_{\text{stego}}^{i+1} - \phi(z^i). \quad (8)$$

For a network with M concealment blocks, during the backward process, the M -th block takes inputs x_{stego} and z , and the outputs are x_{stego}^{M-1} and z^{M-1} , while the outputs of the first block are the ultimately revealed cover audio $x_{\text{cover}}^{\text{rev}}$ and revealed secret $x_{\text{secret}}^{\text{rev}}$.

2) *Noise Layers*: To enhance robustness of the network against various types of distortions, we design a differentiable noise layer, which is inserted between the encoder and decoder during the training process of the network. The differentiable noise layer utilized during training consists of three types of noise sources:

$$N_{\text{pool}} = \{\text{Identity}, \text{Speckle}, \text{Gaussian}, \text{Resample}\}. \quad (9)$$

During training, for each batch of audio, we randomly select one of these noise types from N_{pool} to apply as perturbation.

In order to assess the model robustness to different types of noise, we utilize a noise layer during testing that includes

$$N_{\text{pool}}^{\text{test}} = \{\text{Identity}, \text{Speckle}, \text{Gaussian}, \text{Resample}, \text{MP3 compression}, 8\text{-bit reduction}\}. \quad (10)$$

We provide two versions of the training process: one with the noise layer incorporated and one without. In the training of the model with the noise layer, we build upon a pre-trained model without the noise layer. We decouple the forward and backward processes of the pre-trained model, fixing the parameters of the forward concealment process, while only training the parameters of the backward reveal process. Concretely, after the differentiable noise layer is inserted, the parameters in the forward concealment branch remain frozen throughout the decoupled stage, and only the backward reveal branch is updated. This approach helps ensure that the concealed information remains imperceptible and allows the reveal network to adapt to the presence of noise.

D. Security Mechanism

To ensure the security and integrity of the hidden audio data, our method integrates a scrambling encryption mechanism into the data hiding process. To achieve this, we segment the original secret audio $x_{\text{secret}}^{\text{orig}}$ into very short fragments and then scramble all segments using a pseudo-random number generator (PRNG) π as an encryption measure. The seed σ used for the PRNG is securely shared between the sender and receiver to ensure that only authorized parties can reconstruct the original secret audio.

1) *Encryption Function*: The encryption of $x_{\text{secret}}^{\text{orig}}$ occurs before hiding. We first divide $x_{\text{secret}}^{\text{orig}}$ into n segments:

$$x_{\text{secret}}^{\text{orig}} = [x_1, x_2, \dots, x_n]. \quad (11)$$

Then, the encrypted secret audio x_{secret} can be represented as:

$$x_{\text{secret}} = [x_{\pi(\sigma)(1)}, x_{\pi(\sigma)(2)}, \dots, x_{\pi(\sigma)(n)}], \quad (12)$$

where $\pi(\sigma)(i)$ indicates the new position of the i -th element of the original array after scrambling. This scrambling acts as an encryption step, effectively obfuscating the audio segments before they are embedded into the cover audio.

2) *Decryption Function*: The decryption process is the reverse of the encryption process. First, the encrypted secret audio $x_{\text{secret}}^{\text{rev}}$ is revealed from the stego audio and divided into n segments:

$$x_{\text{secret}}^{\text{rev}} = [x'_1, x'_2, \dots, x'_n]. \quad (13)$$

Subsequently, the original order and content of the secret audio are restored using the inverse of the pseudo-random scrambling sequence used for encryption:

$$x_{\text{secret}}^{\text{restored}} = [x'_{\pi^{-1}(\sigma)(1)}, x'_{\pi^{-1}(\sigma)(2)}, \dots, x'_{\pi^{-1}(\sigma)(n)}], \quad (14)$$

where $\pi^{-1}(\sigma)(i)$ is the inverse mapping of the scrambling, indicating the original position that the i -th element should be restored to after scrambling.

3) *Security Analysis*: To quantify the security of our method, we calculate the probability P that an attacker successfully extracts the secret message. The security of the encryption depends on the unpredictability of the pseudo-random sequence, assuming only the sender and intended receiver know this sequence. Thus, the probability P that an attacker successfully extracts the secret message can be calculated using the number of possible permutations of n segments:

$$P = \frac{1}{n!}. \quad (15)$$

For an audio segment lasting 2 seconds, with $n = 100$ resulting in segment lengths of 20 ms, the key space size is $n! \approx 9.33 \times 10^{157}$, and the probability of the secret being unauthorizedly extracted is 1.07×10^{-158} . In a brute-force attack scenario, the attacker would need to employ automated evaluation tools, such as Automatic Speech Recognition (ASR) systems or Mean Opinion Score (MOS) prediction models, to verify semantic coherence or perceptual quality of each permutation. Even under a conservative assumption where an attacker can validate one permutation every 0.01 s, the expected brute-force search time would still be approximately 1.48×10^{148} years, far exceeding the age of the universe. Furthermore, 20 ms is shorter than a typical English phonetic segment [47], [48], and semantically meaningful words usually consist of one or more phonemes; therefore, such short audio segments are unlikely to convey coherent semantic information. The short duration of audio segments and the computational infeasibility of brute-force attacks ensure that the secret message will not be extracted by unauthorized parties.

Building on this foundation, our security mechanism not only ensures the confidentiality of the hidden data but also its integrity, as any unauthorized attempt to modify or reorder the data without the correct sequence would result in a noticeable degradation of the recovered audio quality, alerting the receiver to potential tampering. Furthermore, the proposed scrambling mechanism effectively decouples detectability from confidentiality: even if the presence of hidden data is statistically detected, the semantic content remains confidential and inaccessible to unauthorized parties. Additionally, the encryption mechanism is designed to be plug-and-play, allowing it to be seamlessly integrated with different segmentations n and keys σ . This flexibility ensures that no additional training or complex configuration adjustments are necessary when changing the number of segments or updating the encryption key, and the computational overhead of scrambling and de-scrambling is negligible compared with the invertible network inference.

E. Loss Function

The overall loss function comprises two components: one part is the guidance loss, intended to encourage the stego audio to resemble the original cover audio and the distribution of the redundant information r to resemble an isotropic spherical Gaussian distribution. Another part is the reconstruction loss, designed to encourage the recovered secret audio and cover audio to resemble the original secret audio and cover audio.

1) *Concealing Loss*: The forward concealment process begins with encrypting the original secret audio $x_{\text{secret}}^{\text{orig}}$ to produce x_{secret} . Subsequently, we generate the stego audio x_{stego} and the redundant information r from the cover audio x_{cover} and the encrypted secret audio x_{secret} . For x_{stego} , we encourage it to be similar to x_{cover} , resulting in the following expression:

$$\mathcal{L}_{\text{stego}}(\theta) = \sum_{n=1}^N \ell_C(x_{\text{cover}}^{(n)}, x_{\text{stego}}^{(n)}), \quad (16)$$

where θ represents the network parameters, and x_{stego} is the output of the function $f_{\theta}(x_{\text{cover}}, x_{\text{secret}})$. The function ℓ_C is used to measure the similarity between x_{cover} and x_{stego} .

For the redundant information r , we aim to make r as close as possible to a specified distribution p , which is independent of the case. This facilitates the recovery stage, where the auxiliary variable z , with a distribution similar to that of the redundant information r , is used to recover x_{secret} in the absence of redundant information r . Therefore, we compute the log-likelihood of the redundant information r conforming to the specified distribution p :

$$\mathcal{L}_z(\theta) = - \sum_{n=1}^N \log(p(r^{(n)})). \quad (17)$$

Without loss of generality, we employ a spherical multivariate Gaussian distribution $p(r) = N(r; 0, I)$ as the specified distribution.

The final loss in the concealment stage is a weighted sum of these two loss components:

$$\mathcal{L}_{\text{conceal}}(\theta) = \lambda_{\text{stego}} \mathcal{L}_{\text{stego}}(\theta) + \lambda_z \mathcal{L}_z(\theta), \quad (18)$$

where λ_{stego} and λ_z are weights used to balance the different loss terms.

2) *Revealing Loss*: The backward recovery process involves the recovery of the cover audio $x_{\text{cover}}^{\text{rev}}$ and encrypted secret $x_{\text{secret}}^{\text{rev}}$ from the stego audio x_{stego} and the auxiliary variable z . Following this, we decrypt $x_{\text{secret}}^{\text{rev}}$ to obtain the secret audio $x_{\text{secret}}^{\text{restored}}$. We encourage $x_{\text{cover}}^{\text{rev}}$ to be similar to x_{cover} and the decrypted secret $x_{\text{secret}}^{\text{restored}}$ to be similar to $x_{\text{secret}}^{\text{orig}}$, leading to the following expressions:

$$\mathcal{L}_{\text{cover}}(\theta) = \sum_{n=1}^N \ell_R(x_{\text{cover}}^{(n)}, x_{\text{cover}}^{\text{rev}(n)}), \quad (19)$$

$$\mathcal{L}_{\text{secret}}(\theta) = \sum_{n=1}^N \ell_R(x_{\text{secret}}^{\text{orig}(n)}, x_{\text{secret}}^{\text{restored}(n)}), \quad (20)$$

where the function $\ell_R(x_1, x_2)$ is employed to gauge the similarity between x_1 and x_2 . The final revealing loss is a weighted sum of these two loss components:

$$\mathcal{L}_{\text{reveal}}(\theta) = \lambda_{\text{cover}} \mathcal{L}_{\text{cover}}(\theta) + \lambda_{\text{secret}} \mathcal{L}_{\text{secret}}(\theta), \quad (21)$$

where λ_{cover} and λ_{secret} are weights used to balance the different loss terms.

3) *Time-Frequency-Domain Loss*: To ensure that the stego and recovered secret audio closely match their respective groundtruth audio in terms of auditory perception and time-frequency structure, we utilize a multi-resolution STFT loss [49]

to bridge the gap between our time-domain architecture and human auditory perception. While modeling in the time domain avoids artifacts caused by transformations and phase inaccuracies, relying solely on time-domain losses often fails to capture perceptual characteristics. This loss serves as a constraint to retain the benefits of frequency-domain modeling for naturalness, while maintaining a purely time-domain pipeline. We denote the set of STFT window sizes used for spectral computation as $W = [w_1, w_2, \dots, w_k]$, where k is the number of window sizes in this set, and w represents the window size used in STFT calculations. Given an STFT window size of m , the amplitude spectrum of the audio is represented as $S^m(\cdot)$. The multi-resolution STFT loss is defined as:

$$\mathcal{L}_{\text{STFT}}(x_1, x_2) = \sum_{w \in W} \|S^w(x_1) - S^w(x_2)\|. \quad (22)$$

To ensure that the outputs of the concealment and reveal networks align with human auditory perception and the original time-frequency structure, we apply the $\mathcal{L}_{\text{STFT}}$ loss function for $\mathcal{L}_{\text{stego}}$, $\mathcal{L}_{\text{cover}}$, and $\mathcal{L}_{\text{secret}}$. During the initial pre-training phase, we use a single-resolution STFT loss with a window size set to $W = [1024]$. After the network has reached convergence, we adjust the window sizes to $W = [256, 512, 1024, 2048]$, enhancing the naturalness of the produced audio.

4) *Total Loss Function*: The overall loss function $\mathcal{L}_{\text{total}}(\theta)$ is the sum of two components, $\mathcal{L}_{\text{conceal}}(\theta)$ and $\mathcal{L}_{\text{reveal}}(\theta)$, and the final total loss function is as follows:

$$\begin{aligned} \mathcal{L}_{\text{total}}(\theta) = & \lambda_{\text{stego}} \mathcal{L}_{\text{stego}}(\theta) + \lambda_z \mathcal{L}_z(\theta) \\ & + \lambda_{\text{cover}} \mathcal{L}_{\text{cover}}(\theta) + \lambda_{\text{secret}} \mathcal{L}_{\text{secret}}(\theta), \end{aligned} \quad (23)$$

where λ_{stego} , λ_z , λ_{cover} , and λ_{secret} are weights used to balance the contributions of the different loss components. Together, these loss terms constrain stego quality, secret recovery, the auxiliary variable, and the multi-resolution STFT consistency.

IV. EXPERIMENTS

In this section, we initially present the datasets used and our experimental setup. Then, we evaluate the performance of our method from both objective and subjective perspectives, comparing it with existing state-of-the-art methods. Objective experiments primarily encompass fidelity, robustness, and generalization aspects, while subjective experiments involve imperceptibility tests. Finally, we investigate the impact of individual components constituting our method.

A. Experimental Setup

1) *Datasets and Settings*: We employ the standard training/validation/testing split and conduct training and testing on well-known speech datasets, specifically the VCTK dataset and the LibriSpeech dataset. The VCTK dataset includes speech data from 110 English speakers with different accents, each reading approximately 400 sentences. To evaluate the generalization of different methodologies across unseen speakers, we earmark speech data from four individuals—specifically, speakers 247, 305, 307, and 374—as the test subset to represent unseen speakers. On the other hand, LibriSpeech consists of 1000 hours of

read English speech. All audio excerpts are uniformly resampled to both 16 kHz and 24 kHz to ensure consistency in our evaluations.

Our training examples are generated by randomly selecting one utterance as the cover and another utterance as the secret. Consequently, the pairing of covers and secrets is not fixed and could involve different speakers. To ensure consistency and uniformity in data handling, all audio is segmented into 2-second clips. In the scrambling encryption process, the parameter n is set to 100, allowing each 2-second audio segment to be divided into 100 parts of 20 ms each. For samples with a 16 kHz sampling rate, the batch size is set to 4, while for samples with a 24 kHz sampling rate, the batch size is set to 2. The number of invertible blocks (M) used in the INN is set to 16. The parameters λ_{stego} , λ_z , λ_{cover} , and λ_{secret} are set to 2.0, 10.0, 1.0, and 1.0, respectively. All models are trained using the Adam optimizer for 80 epochs, with an initial learning rate of $10^{-4.5}$ that is decayed by a factor of 10 every 20 epochs to improve training stability. For the robust decoupled setting, the model trained without the noise layer for 80 epochs is used as the initialization. We then insert the differentiable noise layer, freeze the forward concealment branch, and further train only the backward reveal branch for 30 epochs using Adam with a learning rate of $10^{-4.8}$.

2) *Benchmarks*: To validate the effectiveness of our approach, we compared our method against state-of-the-art (SOTA) audio steganography techniques, which include a traditional 8-bit LSB method, a naïve baseline approach called Frequency Chop, and a deep-learning-based method: Hide & Speak [3]. In Frequency Chop, we concatenated the upper frequency portion of x_{secret} with the lower frequency portion of x_{cover} to create x_{stego} . For a fair comparison, we trained our model and the Hide & Speak [3] model using the same dataset.

3) *Metrics*: To assess the performance of our method in terms of fidelity and robustness, we utilized both objective and subjective metrics to assess the quality of the cover/stego pairs and secret/recovery pairs. The objective metrics employed include Signal-to-noise ratio (SNR), Log-Spectral Distortion (LSD), Perceptual Evaluation of Speech Quality (PESQ), and Speaker Embedding Cosine Similarity (SECS). Higher values of SNR, PESQ, SECS and lower values of LSD suggest superior audio quality. In terms of subjective assessment, we employed ABX test. If the test outcome is close to 50%, this implies that the stego audio and the cover audio are indistinguishable to listeners, thus signifying the imperceptibility of the hidden information.

B. Objective Evaluation

1) *Fidelity*: Table II provides a comparison of the fidelity for our method against other approaches using SNR, LSD, PESQ, and SECS metrics on the LibriSpeech and VCTK datasets. The data clearly demonstrate that our method generally outperforms others in most cases across all metrics for the cover/stego audio pairs, and significantly excels in the secret/recovery audio pairs.

With respect to SNR, our cover/stego audio pairs show enhancements of 0.15 dB, 0.01 dB, and 0.01 dB over the second-best results on the LibriSpeech, VCTK-16 kHz, and

TABLE II
BENCHMARK COMPARISONS ON DIFFERENT DATASETS, WITH THE BEST RESULTS IN BOLD AND THE SECOND BESTS UNDERLINED

Cover/Stego Speech Pair												
Method	LibriSpeech-16 kHz				VCTK-16 kHz				VCTK-24 kHz			
	SNR(dB)↑	LSD↓	PESQ↑	SECS↑	SNR(dB)↑	LSD↓	PESQ↑	SECS↑	SNR(dB)↑	LSD↓	PESQ↑	SECS↑
LSB	3.59	1.88	1.12	0.57	3.44	1.85	1.11	0.52	3.45	1.96	1.08	0.53
Freq. Chop	1.97	1.79	1.12	0.80	2.23	1.78	1.13	0.80	2.28	1.91	1.09	0.84
Hide&Speak	<u>26.88</u>	<u>0.79</u>	<u>3.28</u>	0.96	<u>27.28</u>	<u>0.74</u>	<u>3.14</u>	0.96	<u>27.51</u>	<u>0.78</u>	<u>3.02</u>	<u>0.96</u>
WavInWav	27.03	0.62	3.89	<u>0.95</u>	27.52	0.61	4.00	0.96	27.52	0.65	3.65	0.97

Secret/Recovery Speech Pair												
Method	LibriSpeech-16 kHz				VCTK-16 kHz				VCTK-24 kHz			
	SNR(dB)↑	LSD↓	PESQ↑	SECS↑	SNR(dB)↑	LSD↓	PESQ↑	SECS↑	SNR(dB)↑	LSD↓	PESQ↑	SECS↑
LSB	-10.31	3.36	1.03	<u>0.41</u>	-9.64	3.29	1.03	0.37	-9.60	3.53	1.03	0.40
Freq. Chop	<u>-1.93</u>	1.82	1.67	0.22	<u>-1.89</u>	1.77	<u>1.58</u>	0.17	<u>-1.82</u>	1.73	<u>1.48</u>	0.29
Hide&Speak	-2.60	<u>1.47</u>	1.24	<u>0.41</u>	-2.65	<u>1.40</u>	<u>1.22</u>	<u>0.40</u>	-2.68	<u>1.44</u>	1.21	0.81
WavInWav	11.16	1.32	<u>1.49</u>	0.69	12.86	1.21	1.61	0.79	16.79	1.05	1.70	0.91

VCTK-24 kHz datasets, respectively. Meanwhile, our secret/recovery audio pairs demonstrate substantial improvements of 13.76 dB, 15.51 dB, and 19.47 dB, respectively. It is evident that the traditional LSB method exhibits subpar hiding quality due to its fixed and limited capacity. For example, the LSB and Frequency Chop methods manage SNR values of only 3.59 dB and 1.97 dB for the cover/stego audio pairs, respectively, which are over 20 dB lower than those achieved by our method. Such low SNR levels are easily detectable by the human ear, marking a clear failure in effective audio concealment. Compared to other deep-learning-based methods, our approach achieves markedly superior quality in the recovered secret audio.

The advantages of our method are further evidenced in additional metrics such as LSD, PESQ and SECS. Notably, a higher SECS indicates that the recovered secret audio retains more speaker identity information, showcasing the effectiveness of our method in preserving speaker characteristics within the secret audio. Additionally, our time-domain approach not only outperforms in time-domain metrics like SNR but also demonstrates better performance in frequency-domain metrics like LSD compared to previous frequency-domain methods. This indicates that the time-frequency loss integrated into our model promotes an understanding of the interrelations between the frequency and time domains, enhancing overall model performance.

To further compare speech quality, we visualize the obtained stego audio and recovered secret audio from different methods, as shown in Figs. 5 and 6. It is evident that both our method and Hide & Speak [3] produce stego audio that is indistinguishable from the cover audio. However, in terms of the recovered secret audio, our method exhibits minor distortions, while the recovered secret audio obtained from [3] displays noticeable artifacts and noise. This once again demonstrates the superior fidelity achieved by our method.⁴

2) *Robustness*: In addition to fidelity under noise-free conditions, the performance under noisy conditions is also crucial.

⁴Our demo is available at <https://cyberrange.github.io/project/wavinwav>.

TABLE III
COMPARISON OF RESULTS OBTAINED BY APPLYING DIFFERENT NOISES TO STEGO AUDIO IN THE VCTK DATASET WITH BENCHMARK

	Method	Metrics			
		SNR(dB)↑	LSD↓	PESQ↑	SECS↑
Cover/Stego	Hide&Speak	27.51	0.78	3.02	0.96
	WavInWav	27.52	0.65	3.65	0.97
	WavInWav*	27.52	0.65	3.65	0.97
Identity	Hide&Speak	-2.68	1.44	1.21	0.39
	WavInWav	16.79	1.05	1.70	0.91
	WavInWav*	11.14	1.12	1.48	0.76
Down-Sampling	Hide&Speak	-2.79	1.72	1.10	0.35
	WavInWav	4.42	1.56	1.05	0.46
	WavInWav*	7.67	1.25	1.21	0.49
Gaussian	Hide&Speak	-2.81	2.07	1.07	0.26
	WavInWav	-7.53	2.47	1.04	0.25
	WavInWav*	2.12	1.31	1.06	0.27
Speckle	Hide&Speak	-2.72	1.65	1.14	0.36
	WavInWav	2.27	1.68	1.08	0.67
	WavInWav*	6.79	1.20	1.21	0.58
MP3 128 kbps	Hide&Speak	-2.50	1.52	1.16	0.38
	WavInWav	9.48	1.40	1.18	0.77
	WavInWav*	8.91	1.25	1.27	0.66
MP3 96 kbps	Hide&Speak	-2.58	1.67	1.12	0.35
	WavInWav	4.36	1.57	1.09	0.66
	WavInWav*	6.59	1.30	1.17	0.57
MP3 64 kbps	Hide&Speak	-3.27	2.08	1.07	0.27
	WavInWav	-1.37	1.85	1.05	0.40
	WavInWav*	2.39	1.44	1.07	0.39
MP3 32 kbps	Hide&Speak	-4.49	2.23	1.05	0.14
	WavInWav	-3.91	1.84	1.04	0.00
	WavInWav*	-3.16	1.68	1.04	-0.02
8-bit Reduction	Hide&Speak	-2.87	2.10	1.05	0.23
	WavInWav	-6.03	2.32	1.04	0.23
	WavInWav*	1.62	1.35	1.06	0.26

* The asterisk denotes models with noise layer added during the training process. The best results are indicated in bold.

We applied different channel distortions and compression techniques to the stego audio, and Table III compares the performance of different methods under these distortion conditions, using the noise pool consistent with (10). Gaussian and Speckle noises were applied with $\sigma = 0.001$ and $\sigma = 0.01$, respectively. DownSampling distortion resampled the 24 kHz stego audio to

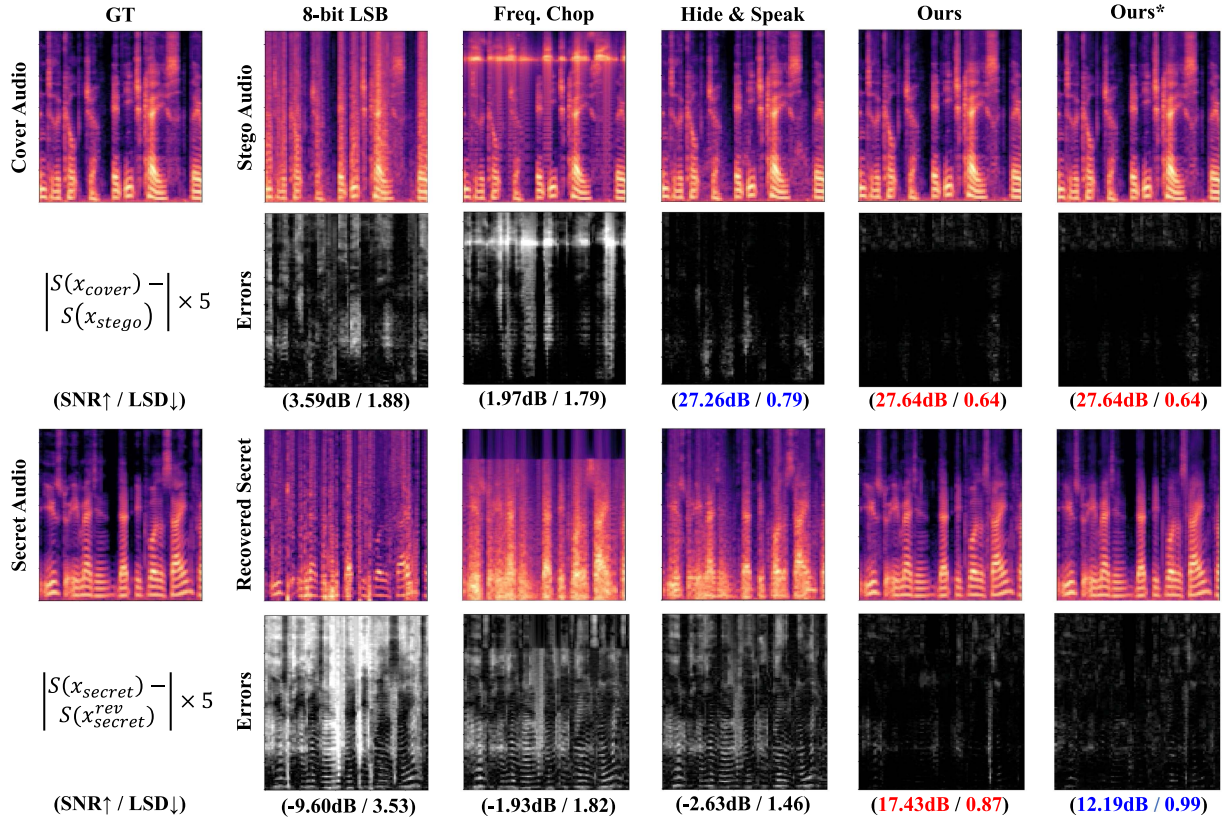


Fig. 5. Visualization of Mel-spectrograms. The hiding operations of LSB, Frequency Chop, Hide&Speak produce visible artifacts on recovered secret audio, whereas our method shows no such artifacts. The asterisk (*) denotes models with noise layer added during training process. Best-performing results are highlighted in red, while second-best results are indicated in blue.

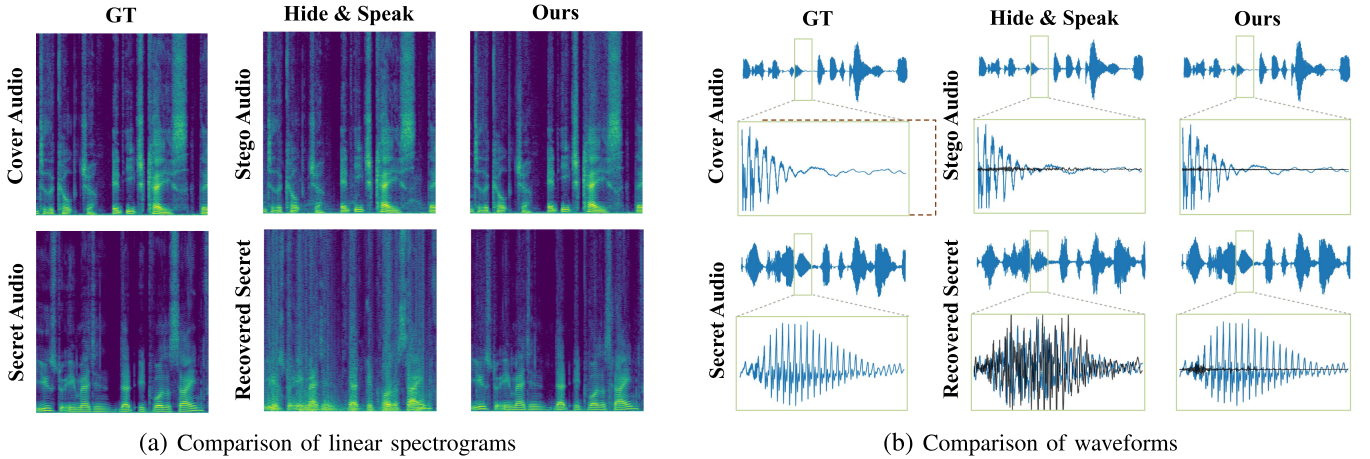


Fig. 6. Visualization of linear spectrograms and waveforms for Cover/Stego pairs (top rows) and Secret/Recovered pairs (bottom rows). To better illustrate the differences in the waveforms, enlarged views are shown below each waveform, wherein the black component represents the residuals.

16 kHz. MP3 compression was applied at rates of 128 kbps, 96 kbps, 64 kbps, and 32 kbps. For our method, we employed two settings: one with and one without the use of the noise pool consistent with (9) during training, denoted as WavInWav* and WavInWav respectively. Since the 8-bit LSB and Frequency Chop baselines have already demonstrated their inability to effectively conceal audio in noiseless conditions, for the sake of

brevity, we omit further comparisons with these two baselines at this stage.

From Table III, it is evident that in most cases, our method outperforms others. Under weaker noise intensity and noise-free conditions, WavInWav shows better performance, whereas under higher noise intensity, WavInWav* performs better, indicating a trade-off between robustness and transparency. Notably,

TABLE IV
APPLICABILITY TO MP3 FORMAT

MP3 Bitrate	Method	Cover Audio Quality				Recovered Secret Quality			
		SNR↑	LSD↓	PESQ↑	SECS↑	SNR↑	LSD↓	PESQ↑	SECS↑
128 kbps	Hide & Speak	27.60	0.83	3.04	0.96	-2.42	1.58	1.18	0.37
	WavInWav	27.61	0.70	3.63	0.97	9.61	1.46	1.19	0.77
	WavInWav*	27.61	0.70	3.63	0.97	8.98	1.30	1.27	0.66
64 kbps	Hide & Speak	27.58	0.83	3.08	0.96	-3.24	2.15	1.07	0.27
	WavInWav	27.38	0.70	3.61	0.96	-1.42	1.93	1.05	0.42
	WavInWav*	27.38	0.70	3.61	0.96	2.41	1.49	1.07	0.39

* The asterisk denotes models with noise layer added during the training process.

WavInWav achieved some robustness against MP3 compression and DownSampling noise even without using the noise layer during training. This is attributed to our use of time-frequency loss during training, enabling the model to learn information hiding and extraction in the frequency domain, while DownSampling and MP3 compression primarily preserve the frequency-domain characteristics of the audio. Additionally, the auxiliary variable z is randomly sampled from a gaussian distribution, contributing to the model robustness.

For MP3 compression below 64 kbps, Gaussian noise, and 8-bit precision reduction, our method exhibited relatively weaker robustness. However, it still outperformed other methods in most metrics. The observed outcomes stem from the intricate balance between robustness and transparency within our method. Our method exhibits minimal alterations in both the frequency and time domains of the cover audio, ensuring a balance between concealing the information and maintaining its fidelity, while the introduction of Gaussian noise and a reduction in precision by 8 bits exert a pronounced influence on the time-domain characteristics of the stego audio. Furthermore, as the bit rate diminishes during MP3 compression, it eliminates a considerable amount of perceptual redundancy, consequently inflicting substantial damage on the imperceptible concealed information. At 32 kbps, the degradation becomes most pronounced, and all compared methods obtain negative SNR values for the secret/recovery audio pairs. Aggressive low-bitrate compression can remove or distort the weak, imperceptible modifications that carry the hidden signal, making secret recovery more difficult at this bitrate. Nevertheless, WavInWav* still achieves the highest secret/recovery SNR among the compared methods at 32 kbps.

3) *Applicability to MP3 Format*: To verify the applicability of our method on different formats, we conducted end-to-end experiments using MP3 format audio as both the cover and secret, and extracted the secret from MP3-compressed stego audio. We evaluated the performance at representative bitrates and compared it with the Hide & Speak baseline [3].

As shown in Table IV, our method maintains high fidelity for the stego audio. For the recovered secret, our method demonstrates advantages over the baseline. For example, our standard method achieves an SNR of 9.61 dB at 128 kbps, and our robust model demonstrates better resilience under more aggressive compression. These findings indicate that our proposed framework is not limited to lossless WAV formats but is effectively applicable to common lossy formats such as MP3.

4) *Generalization Ability*: The generalization performance of audio data hiding methods is crucial for their practical applicability. Our experiments primarily focus on assessing the generalization ability concerning unseen speakers and across different datasets. Specifically, to evaluate the ability on unseen speakers of our model, we isolated four speakers from the VCTK dataset as the unseen speaker set while training on the remaining 106 speakers. For assessing generalization across datasets, we trained the model on the LibriSpeech dataset and tested it on the VCTK dataset. As depicted in Table V, our approach achieves 26.25 SNR for cover/stego pairs and 11.11 SNR for secret/recovery pairs on unseen speakers. Additionally, on the unseen dataset, our method achieved 27.83 SNR for cover/stego pairs and 12.60 SNR for secret/recovery pairs, highlighting its robust generalization performance.

C. Subjective Detectability Test

To verify that humans cannot detect the differences between cover and stego audio, we conducted ABX tests. For each test utterance, we provide two audio samples, A and B, to the volunteers participating in the test. One of these samples is the cover audio, and the other is the stego audio. Subsequently, we present an audio sample X, randomly chosen from A and B, and the participants have to decide whether X is the same as A or B. We generated 20 audio examples, and for each example, we recorded 20 responses, resulting in a total of 400 responses. Only 50.75% of the stego audio could be distinguished from the cover by humans, which is close to random guessing (50%). This suggests that the distortion caused by our method is nearly imperceptible to the human ear.

D. Steganalysis Resistance

To further evaluate the undetectability of the proposed method against steganalysis techniques, we employed two widely recognized deep steganalysis models, LinNet [50] and BSNet [51], and compared our method with Hide & Speak [3]. We simulated a scenario where an attacker collects a limited number of stego audio samples generated by the target method to retrain the detection models. We conducted tests with leaked sample sizes of 100, 200, 500, 1000, 2000, and 4000. Considering potential compression during audio transmission, we evaluated both WAV and MP3 formats. Furthermore, to account for potential differences between the distributions of data collected by an attacker

TABLE V
COMPARISON OF RESULTS OBTAINED BY APPLYING DIFFERENT METHODS TO UNSEEN SPEAKERS AND UNSEEN DATASETS

Method	Cover/Stego Speech Pair						Secret/Recovery Speech Pair					
	Unseen Speakers (VCTK)			Unseen Dataset (VCTK)			Unseen Speakers (VCTK)			Unseen Dataset (VCTK)		
	SNR(dB)↑	LSD↓	SECS↑	SNR(dB)↑	LSD↓	SECS↑	SNR(dB)↑	LSD↓	SECS↑	SNR(dB)↑	LSD↓	SECS↑
Hide&Speak	26.65	0.79	0.95	26.91	0.79	0.96	-2.67	1.42	0.45	-2.60	1.47	0.40
WavInWav	26.25	0.62	0.95	27.83	0.59	0.96	11.11	1.10	0.78	12.60	1.23	0.76

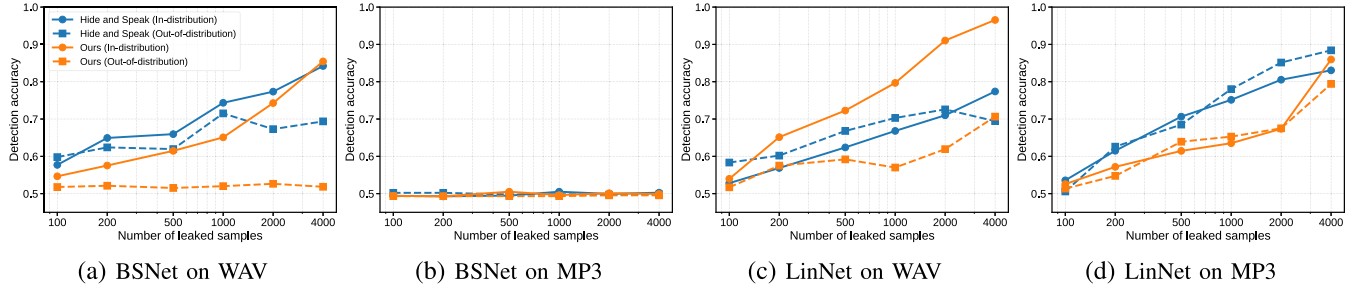


Fig. 7. Detection accuracy of deep learning-based steganalysis models (BSNet and LinNet) on WAV and MP3 formats with varying numbers of leaked training samples. The results compare the proposed method against the Hide & Speak baseline under both in-distribution and out-of-distribution settings.

and the target data, we established two settings: an in-distribution setting (trained and tested on LibriSpeech 16 kHz) and an out-of-distribution setting (trained on LibriSpeech 16 kHz but tested on VCTK 16 kHz). The performance is evaluated using binary detection accuracy, where a value closer to 0.50 indicates lower detectability.

As shown in Fig. 7, the resistance of our method against steganalysis is comparable to the existing method. In the WAV scenario, our method proves more resistant to BSNet compared with the baseline, although LinNet exhibits higher sensitivity to our method. In the MP3 scenario, our method achieves lower detection accuracy against LinNet in most cases, while BSNet performs near random guessing for both methods. Regarding the impact of sample size, detectability for both methods rises as the number of leaked samples increases, consistent with expectations for supervised training. We also observe that detection accuracy for our method increases more notably as the sample size grows. Moreover, our method exhibits lower detectability than the baseline in out-of-distribution settings. Consequently, our method demonstrates a more favorable relative performance against the baseline in the low-sample regime and out-of-distribution settings, which is more realistic when large-scale collection of distribution-matched stego samples is difficult.

E. Computational Complexity

We conducted all experiments on Intel Xeon Gold 6130 CPUs and a single NVIDIA RTX A6000 GPU. Our model contains 3.39 million parameters, with a storage size of approximately 13 MB. In the training phase, we trained on the LibriSpeech train-clean-100 subset for 80 epochs, resulting in a total training time of about 10 days. The peak GPU memory consumption during training was 22.62 GB.

TABLE VI
COMPARISON OF INFERENCE EFFICIENCY

Method	RTF	GPU Mem	System RAM
Hide&Speak	1.0765	2.56 GB	3.66 GB
WavInWav	0.0163	1.14 GB	3.18 GB

Regarding inference efficiency, we first analyze the computational complexity. Theoretically, due to the non-autoregressive nature of our flow-based INN, the complexity scales linearly ($O(T)$) with the audio length T . Quantitatively, the model requires approximately 108 GFLOPs to process 1 s of audio in our experiments.

We further compared the actual inference performance with the baseline Hide & Speak [3] in Table VI. Our method achieves a Real-Time Factor (RTF) of 0.0163, making it suitable for real-time applications. In contrast, the baseline has an RTF of 1.0765, which fails to meet real-time requirements. Furthermore, our end-to-end architecture also consumes less GPU memory usage and system RAM usage compared to the baseline.

F. Ablation Study

1) *Effect of Time-Domain Hiding*: To assess the impact of hiding in different domains, we adapted our model to process the amplitude spectrum by replacing 1D convolutional layers with 2D convolutional layers, ensuring a comparable number of parameters, and retrained the network. As shown in Table VII, hiding audio in the time domain resulted in a significant increase of 13.95 in SNR for cover/stego speech pairs and 15.35 in SNR for secret/recovery speech pairs. This effect stems from directly embedding and extracting in the time domain, which avoids distortions introduced to the stego audio spectrogram by the STFT-iSTFT transformation, thereby preventing the extraction of the secret spectrogram from a distorted stego spectrogram.

TABLE VII
ABLATION STUDY ON INVERTIBLE NEURAL NETWORK, TIME-DOMAIN HIDING, TIME-FREQUENCY-DOMAIN LOSS AND SUB-NETWORK ARCHITECTURE

Sub-block Arch.	INN	Time-domain Hiding	Time-freq. Loss	Cover/Stego Speech Pair				Secret/Recovery Speech Pair			
				SNR(dB)↑	LSD↓	PESQ↑	SECS↑	SNR(dB)↑	LSD↓	PESQ↑	SECS↑
1D-Dense CNN	✗	✓	✓	24.65	0.76	3.40	0.93	11.68	1.27	1.47	0.73
1D-Dense CNN	✓	✗	✓	13.57	0.92	1.54	0.71	-2.49	1.28	1.11	0.23
1D-Dense CNN	✓	✓	✗	26.84	0.83	2.83	0.94	12.25	1.49	1.26	0.70
Transformer	✓	✓	✓	20.13	1.13	1.81	0.92	6.94	1.67	1.11	0.47
1D-Dense CNN	✓	✓	✓	27.52	0.61	4.00	0.96	12.86	1.21	1.61	0.79

Additionally, since the extracted secret audio is also in the time domain, there is no need to infer the phase spectrum from the amplitude spectrum, further circumventing severe phase distortions resulting from inferring the phase from the distorted amplitude spectrum.

2) *Effect of INN*: As an alternative to the INN architecture, we independently modeled the hiding and extraction processes using the traditional encoder-decoder framework. Specifically, we ensured a similar quantity of network parameters and constructed two U-Net architectures, akin to our experimental setup, as separate networks for the hiding and extraction processes. From Table VII, we observed that the INN-based network demonstrated superior overall performance. We attribute this to the intrinsic reversibility of the INN, which naturally fits the modeling of the inverse processes involved in information hiding and extraction. Hence, the utilization of INN yielded improved fidelity.

3) *Effect of Sub-Network Architecture*: We compared the 1D-CNN dense blocks with a transformer block [52] of comparable parameter size, utilizing non-overlapping waveform patches as tokens. As shown in Table VII, our CNN-based method outperforms the transformer variant, confirming the fine-grained local modeling provided by CNNs is essential for time-domain audio hiding.

4) *Effect of Time-Frequency-Domain Loss*: In Table VII, we conducted a comparative analysis against ℓ_2 time-domain loss. The outcomes revealed that the time-frequency loss contributed to an increase of 0.68 in SNR for cover/stego speech pairs and 0.61 in SNR for secret/recovery speech pairs. Moreover, there were more pronounced improvements in LSD and PESQ metrics, with a 1.17 enhancement in PESQ for cover/stego speech pairs and a 0.35 enhancement for secret/recovery speech pairs. As LSD primarily reflects time-frequency characteristics and PESQ demonstrates a stronger alignment with time-frequency metrics than time-domain metrics, these outcomes align intuitively. However, the time-frequency loss also engendered enhancements in the time-domain metric SNR. This phenomenon might stem from the challenge posed by the time-domain loss, making it harder for the model to capture the time-frequency correlations in the audio, thereby impeding convergence to the optimal solution.

5) *Effect of Decoupled Training*: We compared our decoupled training strategy in robust setting with joint training, where forward and backward passes share weights and are updated simultaneously. As shown in Table VIII, while joint training

TABLE VIII
COMPARISON OF DECOUPLING TRAINING AND JOINT TRAINING

		Method	Metrics			
			SNR(dB)↑	LSD↓	PESQ↑	SECS↑
Secret/Recovery	Cover/Stego	WavInWav†	27.00	0.43	3.16	0.94
		WavInWav*	27.52	0.65	3.65	0.97
	Identity	WavInWav†	9.33	1.15	1.35	0.47
		WavInWav*	11.14	1.12	1.48	0.76
	Down-Sampling	WavInWav†	8.56	1.16	1.27	0.44
		WavInWav*	7.67	1.25	1.21	0.49
	Gaussian	WavInWav†	5.98	1.21	1.14	0.34
		WavInWav*	2.12	1.31	1.06	0.27
	MP3	WavInWav†	9.16	1.15	1.33	0.46
	128 kbps	WavInWav*	8.91	1.25	1.27	0.66

Note: The dagger (†) denotes jointly trained models in the robust setting (updating shared weights simultaneously). The asterisk (*) denotes models using our decoupling strategy (freezing the forward pass).

improves robustness against extreme distortions (e.g., higher SNR under Gaussian noise), it comes at the cost of significantly degrading the cover audio quality, where PESQ drops from 3.65 to 3.16. This degradation occurs because the shared weights tightly couple the forward and backward processes; therefore, adapting to noise in the backward pass degrades the forward hiding performance. We also observed distinct audible artifacts in the stego audio generated by joint training, which aligns with the degradation in objective metrics. Since imperceptibility is the prerequisite for data hiding, we adopt the decoupled training strategy as our robust setting.

V. CONCLUSION

In this paper, we introduce WavInWav for the end-to-end concealment and recovery of complete secret audio. We utilize a flow-based invertible neural network to establish a direct mapping between stego, cover, and secret signals. To optimize quality, we employ a time-frequency loss on time-domain signals, while incorporating encryption technology to protect the hidden data from unauthorized access. The combination of time-domain hiding and time-frequency loss effectively reduces the distortions typically caused by time-frequency transformations in previous methods, while retaining the benefits of time-frequency constraints for naturalness.

Qualitative results show that modifications to the cover audio are imperceptible to human listeners. Quantitatively, our method significantly outperforms state-of-the-art methods in terms of audio fidelity, notably achieving SNR improvements ranging

from 13 to 20 dB for the recovered secret audio. Additionally, the method demonstrates robustness against multiple types of noise and generalization to unseen speakers and datasets, and resistance against steganalysis comparable to existing methods.

However, there are still limitations. Our method is vulnerable to strong noise attacks and severe low-bitrate MP3 compression, both of which can affect minor modifications to the stego audio. In terms of undetectability, although our method demonstrates favorable relative performance in limited-data scenarios, it remains vulnerable to steganalysis given large-scale training data. To further improve undetectability, future work could explore a multi-resolution STFT steganalyzer as an adversarial discriminator, where the steganalyzer could provide an adversarial signal that makes stego audio harder to distinguish from cover audio while preserving secret recovery. Finally, extending the framework to various audio formats and other domains like video hiding remains a promising direction.

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Xiangkun Wang received the BS degree from the University of Science and Technology of China (USTC) in 2021. He is currently working toward the graduation degree with USTC. His research interests include steganography and deep learning.



Weiming Zhang received the MS degree and PhD degree from the Zhengzhou Information Science and Technology Institute, China, in 2002 and 2005, respectively. He is currently a professor with the School of Information Science and Technology, University of Science and Technology of China. His research interests include information hiding and multimedia security.



Wei Fan received the BS degree from the University of Science and Technology of China (USTC) in 2022. He is currently working toward the graduation degree with USTC. His research interests include information hiding and deep learning.



Nenghai Yu received the BS degree from the Nanjing University of Posts and Telecommunications in 1987, the ME degree from Tsinghua University in 1992, and the PhD degree from the University of Science and Technology of China in 2004. He is currently a professor with the University of Science and Technology of China. His research interests include multimedia security, multimedia information retrieval, video processing, and information hiding.



Kejiang Chen (Senior Member, IEEE) received the BS degree from Shanghai University in 2015 and the PhD degree from the University of Science and Technology of China (USTC) in 2020. He is currently an associate research fellow with USTC. His research interests include information hiding, image processing, and deep learning.